

Chief Scientist Office 2026 Team Description Paper

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<https://www.robocup.ros-sier.com/>, <https://github.com/sbgisen>

Abstract This paper presents SoftBank Corp. Japan team Chief Scientist Office’s mobile manipulator “SOAR” for RoboCup 2026 @Home Open Platform League.

1 Introduction

This paper presents the continued development of SOAR, a mobile manipulator robot designed by SoftBank Corp. Chief Scientist Office for participation in RoboCup @Home 2026. Building upon previous iterations, we enhanced system integration, migrating to ROS2 and introducing a robust perception-to-action pipeline that includes but is not limited to 3D scene fitting, object recognition, speech interaction, and behavior planning using large language models. Our research emphasizes how these components coordinate within a unified architecture to enable adaptive planning, robust control, and flexible interaction in dynamic environments. We also report our increasing contributions to the robotics community through open-source source releases and conference sponsorships, and discuss the broader application of SOAR as a customizable and affordable robot platform for research and commercial use.

SOAR occupies a distinct and deliberate role as an internal research platform. Its objective is not productization, but the accumulation of technical knowledge, development experience, and system integration capability that can serve as the foundation for the next generation of SoftBank robots.

The mobile manipulator robot is a long-term investment in robotic intelligence. It functions as a testbed for advance perception, planning, and interaction capabilities - particularly in integrating AI-driven reasoning into physical systems. This focus enables engineers to experiment bottom-up, verify cross disciplinary concepts, and consolidate practical know-how in robotic autonomy without the operational constraints of product development. SOAR thus represents SoftBank’s commitment to internal competence building, emphasizing sustainable research value over short-term deliverables.

As a research platform, SOAR continues to evolve both technically and strategically. Hardware remains stable, but its software stack has undergone a

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major transformation: migrating ROS2 for long-term maintainability, integration of scene-fitting perceptipon pipeline, deployment of new object recognition, and introducing LLM-based behavior planner. These elements collectively position SOAR as an experimental platform for studying how modern AI systems can enhance robotic adaptability and cognitive control in real-world environments such as RoboCup@Home.

1.1 Focus of research

Our research is motivated by the ambition to develop a customizable and affordable robot platform. We recognize the constraints of current technologies and are committed to exploring their limits and potential. Key areas of our interest include but are not limited to:

- Modularity and Configurability: Investigate how to design a system that allows for easy integration and customization of hardware modules.
- Device Integration: Tackling the hardware challenges related to the seamless integration of diverse devices and components

We believe that these research areas are crucial to bridging the gap between the aspirational and the practical, laying the groundwork for a robot that can adapt to various needs.

2 Hardware Modules

SOAR is a mobile manipulator comprising four main components: the mobile base, the torso, the manipulator, and the head as shown in Fig. 1. The manipulator are commercially available products, while the mobile base, the torso, and the head are custom-developed by our team. The head and the torso are separate components designed for specific functionalities; we designed the head to accommodate various interchangeable sensors and the torso to enhance the manipulator’s reach and versatility. All components are designed with minimal cable protrusion, with circuits centralized on the torso, thus streamlining the connection across components with minimal cabling. This modular approach not only facilitates customization but also organizes and simplifies maintenance and upgrades, allowing for the interchange of components as needed.

2.1 Mobile Base

SOAR’s mobile base (Fig. 1b) is a compact 2-wheel differential drive platform, designed for indoor navigation. Both the driving wheels and casters are equipped with suspensions to ensure smooth movement and stability. The mobile base integrates a 2D liDAR and a 9-axis IMU for localization and autonomous navigation. It also includes battery management system and a docking charging connector, enabling the SOAR to recharge itself without human intervention



Figure 1: SOAR and its four components.

thus extending its operational time. The motor drivers and the battery management system are connected to a USB hub, resulting in only two cables extending out: the USB cable from the device hub and the power supply cable.

Previously, SOAR used a commercially available mobile base with custom modifications. For RoboCup @Home 2026, we swapped with a newly developed, in-house base that has smaller footprint and equipped with a built-in LED button for direct user interface.

2.2 Torso

The torso (Fig. 1c), developed by our team, is a one degree of freedom linear actuator designed to vertically adjust the position of the head and manipulator. It incorporates a constant-force spring to provide gravity compensation, reducing motor load and improving energy efficiency during vertical movements. This mechanism provides an addition 350 mm of vertical clearance, significantly enhancing the manipulator’s reach and versatility. The torso also serves as the central hub for SOAR’s hardware components, housing the main PC and controller box.

2.3 Manipulator

The manipulator (Fig. 1d) is xArm 7¹, a commercially available product developed by uFactory. The end-effector is equipped with a two-finger gripper and a RGB-D hand camera.

2.4 Head

The head (Fig. 1e), designed as SOAR’s primary sensory and interactive module, serves as a centralized platform for perception and user engagement. The module supports various range of sensors and user interface devices for environmental understanding and social interaction.

Equipped with a USB hub for efficient cable management and a separate USB for the high-bandwidth depth camera, it allows for easy swapping of sensors and devices, ensuring that the component can be customized for specific requirements. The head extends three specific cables: a USB cable from the sensor hub, a separate USB for the depth camera, and a power supply cable.

In addition, the head has 2 degrees of freedom for pan and tilt movements, allowing SOAR to dynamically adjust its field of view for environment scanning, object tracking, and social interaction.

3 Software

3.1 Object Recognition

Object recognition is a crucial aspect of SOAR’s perception system, enabling it to identify and interact with various objects in its environment. To achieve

¹ <https://www.ufactory.cc/xarm-collaborative-robot/>



(a) Detecting cup noodles with YOLOv8. (b) Point cloud data and MoveIt planning scene. (c) Generated 3D collision boxes.

Figure 2: Object detection pipeline. YOLOv8’s bounding box (Fig. 2a) and the point cloud data (Fig. 4a) are given to `jsk_pcl_ros` package to generate 3D collision boxes (Fig. 4b), which are passed to MoveIt planning scene.

robust and reliable perception, we combine 2D image-based object detection with 3D point cloud processing. We employed ultralytics’s YOLOv8[1] for real-time object detection, producing labeled bounding boxes for all recognized objects within the robot’s field of view. These bounding boxes are then used to segment and cluster point cloud data within the detected regions to compute 3D collision boxes and object poses, which are essential for downstream manipulation tasks. The pipeline is illustrated in Fig. 2.

3.2 Speech Recognition And Interaction

Our previous system relied on EfficientWord-Net for hotword detection, Google Dialogflow for conversational understanding, and Google Text-To-Speech for vocal responses. Our current system mainly uses ChatGPT [2], known for its advanced natural language processing capabilities, to handle speech transcription, conversational understanding, and speech generation. The speech pipeline begins with Silero VAD [3] to segment incoming audio, followed by GPT-4o mini Transcribe [?] for speech-to-text conversion. The transcribed text is then processed by GPT-4.1 mini [2] dedicated to conversational understanding, which to generate an appropriate response. The responses are converted back into audio using GPT-4o mini TTS.

In addition, we are exploring the use of the ChatGPT Realtime API, which supports low-latency, streaming interactions, and dynamic interaction where responses can be interrupted, updated, or redirected mid-generation based on new user input. To support this real-time conversational loop, we used Google’s WebRTC AE3 echo canceller [?] to suppress audio feedback from the robot’s own speakers, preventing them from being captured by the microphone. This enables continuous, overlapping dialogue where the user can speak while the robot is talking, resulting in more natural and fluid human-robot interaction.

3.3 Behavior

SOAR’s behavior system is built on the SMACH ROS package, which provides a hierarchical state machine framework forming the core of the robot’s decision-making architecture. Each SMACH state is designed as a modular, reusable component, allowing behaviors to be composed, extended, and reorganized for various tasks.

In standard task execution, state transitions follow predefined logic, but for the GPSR (General Purpose Service Robot) task, we use ChatGPT as a task planner to dynamically generate behaviors based on natural-language commands. Given a natural-language command, ChatGPT generates a sequence of states to execute, which are then executed in order without relying on the traditional state transition conditions. This approach enables flexible task planning and adaptation to open-ended commands. For more controlled execution, certain states themselves encapsulate internal state machines, allowing for structured sub-behaviors within a larger LLM-defined plan. As a result, some states function as primitive actions, while others manage complex behaviors, with the task planning model determining their order of execution while the robot’s internal logic governs how each behavior is executed.

A key feature of our current architecture is the use of two separate GPT-4.1 mini models with distinct prompting strategies to clearly separate conversational interaction from task planning. All user input is first processed by the conversational model, which interprets intent, maintains dialogue context, and determines whether the user’s request requires the robot to act. If an action is required, the conversational model passes the request to the task-planning model, which generates a sequence of SMACH states to execute the task.

The behavior pipeline along with its speech interaction for the GPSR task is illustrated in Fig. 3.

3.4 Scene fitting

In household environments, the scene around the robot is rarely static. While certain structures such as walls or large furniture remain fixed, many surfaces, such as tables and shelves where manipulation tasks occur, often shift slightly due to human activities. These subtle changes can introduce significant risks during manipulation, as the mismatch between the robot’s internal scene model and the actual environment can lead to collisions. To address this, we developed a scene fitting process that aligns its internal scene model with the current environment before performing manipulation tasks.

During the mapping phase, SOAR records the point cloud and the pose of key furniture surfaces involved in manipulation tasks, such as tables and shelves. At runtime, the robot captures the point cloud of the same surface and applies a multi-stage ICP alignment to register the current observation with the previously stored model. The resulting transformation is used to update the robot internal scene model pose, ensuring that the motion planning accounts for slight environmental changes.

Fig. 4 illustrates an example of scene fitting for a table.

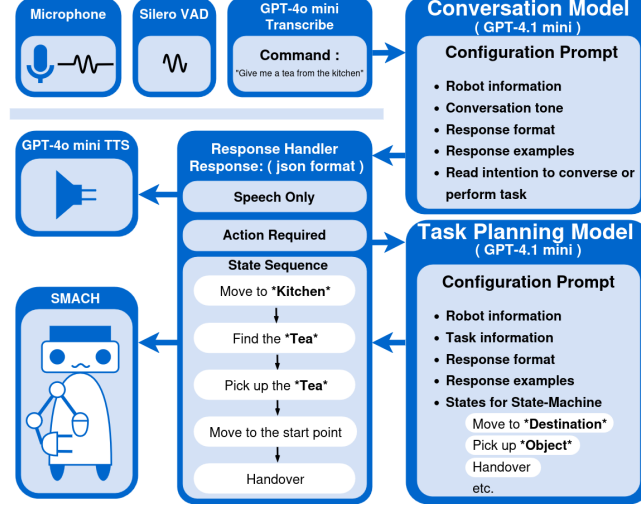
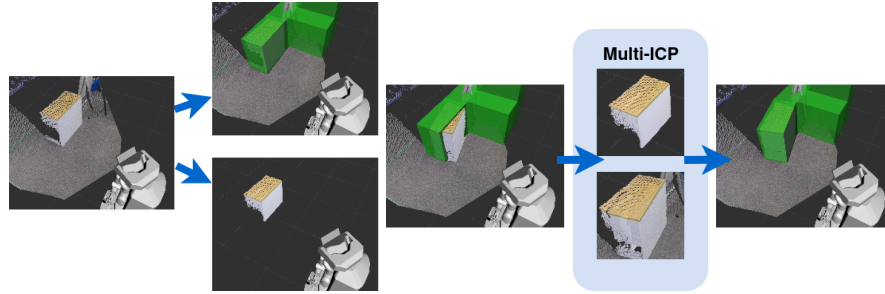


Figure 3: Behavior for GPSR task.



(a) During setup, we create table's (b) When table pose shifts, the robot performs collision box and crop point cloud alignment between the reference model (top) as reference model. and the current environment (bottom) and updates the table's collision box pose.

Figure 4: Example of scene fitting a table.

4 Contribution

Our recent contributions extend beyond the development of SOAR itself, focusing on fostering community growth and supporting research in robotics. We have focused on increasing our impact on the robotics community through open source software contribution and supporting research community initiatives. The following are some of our recent contributions:

- Open-source software modules. We developed and released an open-source hardware interface for VESC motor controller, enabling integration with ROS2 control framework.
- ROS contributions. Members of our team actively contributed to official ROS repositories, including bug fixes, and feature enhancements.
- Robotics conference and competition sponsorships. We recognize the advancing robotics requires more than individual project development and thus we contributed to the growth of the research and educational ecosystem.

5 Conclusions and future work

In this paper, we presented our currently developing robot SOAR which will participate in the RoboCup@Home League. We detailed the hardware specification and introduced several software modules, such as object recognition and speech recognition, which we will continue to refine to tackle the tasks in RoboCup@Home for our future work. By enhancing SOAR’s capabilities in real-world interaction, we aim to contribute not just a flexible and affordable platform but also a robust platform for research and commercial use.

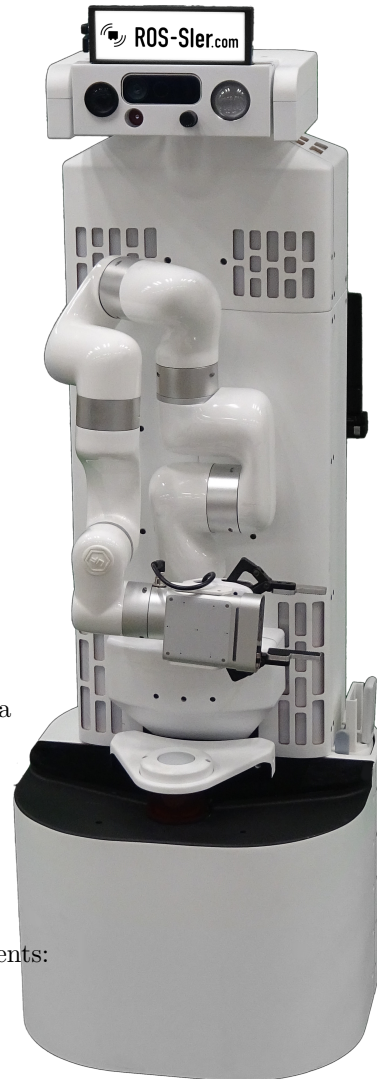
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SOAR Hardware Description

Specifications are as follows:

- Dimensions: (w) 500 mm (d) 530 mm (h) 1440 mm - 1790 mm
- Weight: 113 kg
- Max Speed: 1.2 m/s
- Battery: 25.6V 40Ah LiFePO4, 3 hours operation time
- Mobile Base: differential drive
 - Sensors:
 - * Hinson LE-50621 LiDAR
 - * LDROBOT LD06
 - * Wheeltec N100 9-axis IMU
- Torso unit: 1 DOF linear joint
 - PC: Intel NUC (NUC11PHKi7C)
- Manipulator: 7 DOF, 3.5 kg payload
 - Sensors:
 - * ORBBEC Femto Bolt
- Head unit: 2 DOF pan-tilt
 - Sensors:
 - * Azure Kinect DK RGB-D camera
 - * NEEWER NW-410 condenser microphone
 - * GENIUSPY JM800S-30-360X zoom camera
 - * Seek Thermal CompactPRO Fast Frame
 - Display: THANKO C-79D21B
 - Speaker: YAMAHA VXS1MLB



SOAR Software Description

SOAR use the following open-source software components:

- Platform: Ubuntu 24.04 ROS2 Jazzy
- Navigation: Nav2, MPPI controller
- Manipulation: MoveIt2, MoveIt Task Constructor
- Object Recognition: YOLOv8
- Behavior Architecture: SMACH
- Speech Recognition: Silero VAD, Whisper
- Speech Generation: GPT-4o-mini-tts
- LLM: GPT-4o